

- 1 *Saccharomyces cerevisiae* is a type of yeast. It is a single-celled organism which is widely used in brewing and baking.

The photograph shows yeast cells seen through a microscope.



(Source: © Kateryna Kon/Shutterstock)

- (a) Most of the ATP produced when yeast respires aerobically comes from oxidative phosphorylation in the mitochondria.

Oxidative phosphorylation requires a source of hydrogen ions and the movement of electrons.

- (i) State the location in the mitochondria where oxidative phosphorylation takes place.

(1)

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- (ii) Name **one** molecule that carries hydrogen for use in oxidative phosphorylation.

(1)

(b) Describe an experiment to investigate the effect of temperature on the rate of respiration in yeast, using an artificial hydrogen carrier (redox indicator).

(5)

[illegible]

(c) Explain why temperature affects the rate of respiration in yeast.

(3)

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(Total for Question 1 = 10 marks)
